Week12

Candice Lei (A18585167)

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Table of contents

Population Scale Analysis

> Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```r
url <- "https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.txt'
expr_data <- read.table(url, header = TRUE)

head(expr_data)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```r
summary(expr_data)
```

```
    sample              geno                exp
 Length:462         Length:462         Min.   : 6.675
 Class :character   Class :character   1st Qu.:20.004
 Mode  :character   Mode  :character   Median :25.116
```

```
                                       Mean   :25.640
                                       3rd Qu.:30.779
                                       Max.   :51.518
```

```r
expr_data$geno <- factor(expr_data$geno, levels = c("A/A", "A/G", "G/G"))

table(expr_data$geno)
```

```
A/A A/G G/G
108 233 121
```

```r
tapply(expr_data$exp, expr_data$geno, median)
```

```
     A/A      A/G      G/G
31.24847 25.06486 20.07363
```

```r
# Sample size per genotype
table(expr_data$geno)
```

```
A/A A/G G/G
108 233 121
```

```r
# Median expression per genotype
tapply(expr_data$exp, expr_data$geno, median)
```

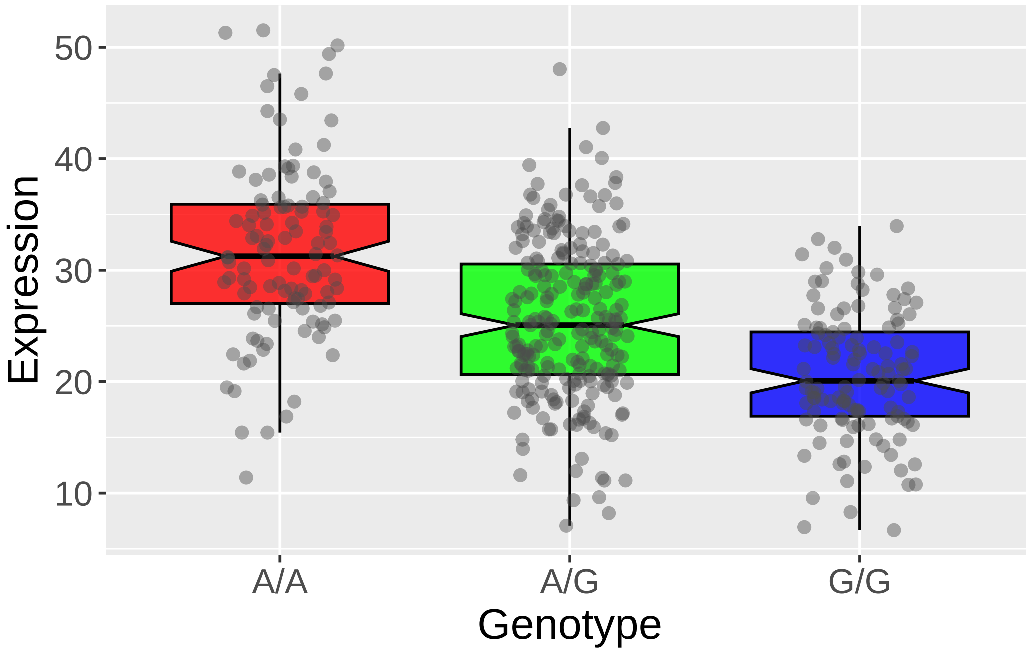
```
     A/A      A/G      G/G
31.24847 25.06486 20.07363
```

> Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```r
library(ggplot2)

ggplot(expr_data, aes(x = geno, y = exp, fill = geno)) +
  geom_boxplot(
    notch = TRUE,
```

```
    outlier.shape = NA,
    alpha = 0.8,
    color = "black"
  ) +
  geom_jitter(
    width = 0.2,
    alpha = 0.45,
    size = 1.8,
    color = "gray30"
  ) +
  scale_fill_manual(
    values = c("A/A" = "red",
               "A/G" = "green",
               "G/G" = "blue")
  ) +
  labs(
    x = "Genotype",
    y = "Expression"
  ) +
  theme_gray() +
  theme(
    legend.position = "none",
    axis.title = element_text(size = 16),
    axis.text = element_text(size = 13)
  )
```


50
40
30
20
10
Expression
A/A
A/G
G/G
Genotype